

NATIONAL BOARD OF ACCREDITATION

Data Capturing Points of the Program Applied for NBA Accreditation– Tier I/II UG (Engineering) Institute Programs

Program Name : Biotechnology	Discipline : Engineering & Technology
Level : Under Graduate	Tier : 1
Application No : 11860	Date of Submission : 20-05-2026

PART A- Profile of the Institute

A1.Name of the Institute : KLE Technological University	
Year of Establishment : 1947	Location of the Institute: BVB Campus, Vidyanagar, Hubballi
A2. Institute Address :KLE Technological University B.V. Bhoomaraddi College Campus , Vidyanagar Hubballi -580031	
City:Dharwad	State:Karnataka
Pin Code:580031	Website:www.kletech.ac.in
Email:vc@kletech.ac.in	Phone No(with STD Code):0836-2378102
A3. Name and Address of the Affiliating University (if any):	
Name of the University :	City:
State :	Pin Code: 0
A4. Type of the Institution : University	
A5. Ownership Status : Self financing	

A6. Details of all Programs being Offered by the Institution:

- No. of UG programs: 11
- No. of PG programs: 8

Table No. A6.1: List of all programs offered by the Institute.

Sr.No.	Discipline	Level of program	Name of the program	Year of Start	Year of Closed	Name of The Department
1	Architecture	UG	Architecture	2015	--	Architecture
2	Computer Application	PG	Master of Computer Application	2015	--	Computer Application
3	Computer Application	PG	Master of Computer Applications (Integrated)	2024	--	Computer Application
4	Engineering & Technology	PG	Advanced Manufacturing Systems	2020	--	Mechanical Engineering
5	Engineering & Technology	UG	Automation & Robotics	2015	--	Automation and Robotics
6	Engineering & Technology	UG	Biotechnology	2015	--	Biotechnology
7	Engineering & Technology	UG	Civil Engineering	2015	--	Civil Engineering
8	Engineering & Technology	UG	Computer Science and Engineering	2015	--	Computer Science and Engineering
9	Engineering & Technology	PG	Computer Science and Engineering	2015	--	Computer Science and Engineering

10	Engineering & Technology	UG	Computer Science and Engineering (Artificial Intelligence)	2021	--	Computer Science and Engineering (Artificial Intelligence)
11	Engineering & Technology	PG	Design Engineering	2020	--	Mechanical Engineering
12	Engineering & Technology	UG	Electrical & Electronics Engineering	2015	--	Electrical and Electronics Engineering
13	Engineering & Technology	UG	ELECTRONICS AND COMMUNICATION ENGINEERING	2015	--	Electronics and Communication Engineering
14	Engineering & Technology	UG	Electronics and Communication Engineering (Industry Integrated)	2020	--	Electronics and Communication Engineering
15	Engineering & Technology	UG	Electronics Engineering (VLSI Design and Technology)	2022	--	Electronics Engineering (VLSI Design and Technology)
16	Engineering & Technology	UG	Mechanical Engineering	2015	--	Mechanical Engineering
17	Engineering & Technology	PG	Structural Engineering	2015	--	Civil Engineering
18	Engineering & Technology	PG	VLSI Design & Embedded Systems	2015	--	Electronics and Communication Engineering
19	Management	PG	Master of Business Administration	2015	--	Management

A7. Programs to be considered for Accreditation vide this Application:

Table No. A7.1: List of programs to be considered for accreditation.

Name of the Department	Having Allied Departments	Name of the Program	Program Level
Biotechnology	No	Biotechnology	UG
Electronics and Communication Engineering	Yes	ELECTRONICS AND COMMUNICATION ENGINEERING	UG
Computer Science and Engineering	Yes	Computer Science and Engineering	UG

Table No. A7.2: Allied Department(s) to the Department of the program considered for accreditation as above.
Cluster ID. Name of the Department (in table no. A7.1) Name of allied Departments/Cluster (for table no. A7.1)

No Record

PART-B: Program information**B1. Provide the Required Information for the Program Applied For:**

Table No. B1: Program details.

A. List of the Programs Offered by the Department:

SR.NO.	PROGRAM NAME	PROGRAM APPLIED LEVEL	YEAR OF START / YEAR OF CLOSED	SANCTIONED INTAKE	INCREASE/DECREASE INTAKE (if any)	YEAR OF INCREASE/DECREASE	CURRENT INTAKE	YEAR OF AICTE APPROVAL	AICTE/COMPETENT AUTHORITY APPROVAL DETAILS	ACCREDITATION STATUS	FROM	TO	NO. OF TIMES PROGRAM ACCREDITED	PROGRAM DURATION
1	Biotechnology	UG	2015 / --	60	No	NA	60	2015	AICTE PID: 1-3626724751	Granted accreditation for 3 years for the period (specify period)	2023	2026	2	4

List of the Allied Departments/Cluster and Programs:

B2. Detail of Head of the Department for the program under consideration:

A. Name of the HoD :	Dr. B. S. Hungund
B. Nature of appointment:	Regular
C. Qualification:	Ph.D

B3. Program Details

Table No.B3.1: Admission details for the program excluding those admitted through multiple entry and exit points.

Item (Information to be provided cumulatively for all the shifts with explicit headings, wherever applicable)	2025-26 (CAY)	2024-25 (CAYm1)	2023-24 (CAYm2)	2022-23 (CAYm3)	2021-22 (CAYm4)	2020-21 (CAYm5)	2019-20 (CAYm6)
N=Sanctioned intake of the program (as per AICTE /Competent authority)	60	60	60	60	60	60	60
N1=Total no. of students admitted in the 1st year minus the no. of students, who migrated to other programs/ institutions plus no. of students, who migrated to this program	57	57	53	43	50	50	52
N2=Number of students admitted in 2nd year in the same batch via lateral entry including leftover seats	0	0	1	2	1	0	0
N3=Separate division if any	0	0	0	0	0	0	0
N4=Total no. of students admitted in the 1st year via all supernumerary quotas	0	0	0	0	0	0	0
Total number of students admitted in the program (N1 + N2 + N3 + N4) - excluding those admitted through multiple entry and exit points.	57	57	54	45	51	50	52

CAY= Current Academic Year. CAYm1= Current Academic Year Minus 1 CAYm2= Current Academic Year Minus 2. LYG= Last Year Graduate. LYGm1= Last Year Graduate Minus 1. LYGm2= Last Year Graduate Minus 2.

B4. Enrolment Ratio in the First Year

Table No. B4.1: Student enrolment ratio in the 1st year.

Year of entry	N (From Table 4.1)	N1 (From Table 4.1)	N4 (From Table 4.1)	Enrollment Ratio [(N1/N)*100]
2025-26 (CAY)	60	57	0	95.00
2024-25 (CAYm1)	60	57	0	95.00

2023-24 (CAYm2)	60	53	0	88.33
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Average [(ER1 + ER2 + ER3) / 3] = 92.78≡ 20.00

B5. Success Rate of the Students in the Stipulated Period of the Program

Table No.B5.1: The success rate in the stipulated period of a program.

Item	(2021-22) LYG	(2020-21) LYGm1	(2019-20) LYGm2
A*= (No. of students admitted in the 1st year of that batch and those actually admitted in the 2nd year via lateral entry, plus the number of students admitted through multiple entry (if any) and separate division if applicable, minus the number of students who exited through multiple entry (if any).	61.00	60.00	60.00
B=No. of students who graduated from the program in the stipulated course duration	42.00	47.00	47.00
Success Rate (SR)= (B/A) * 100	68.85	78.33	78.33

Average SR of three batches ((SR_1+ SR_2+ SR_3)/3): 75.17

B6. Academic Performance of the First-Year Students of the Program

Table No.B6.1: Academic Performance of the First-Year Students of the Program.

Academic Performance	CAYm1(2024-25)	CAYm2(2023-24)	CAYm3 (2022-23)
Mean of CGPA or mean percentage of all successful students(X)	7.60	7.29	7.21
Y=Total no. of successful students	54.00	48.00	43.00
Z=Total no. of students appeared in the examination	57.00	52.00	43.00
API [X*(Y/Z)]	7.20	6.73	7.21

Average API[(AP1+AP2+AP3)/3] : 7.05

B7: Academic Performance of the Second Year Students of the Program

Table No.B7.1: Academic Performance of the Second Year Students of the Program.

Academic Performance	CAYm1 (2024-25)	CAYm2 (2023-24)	CAYm3 (2022-23)
X=(Mean of 2nd year grade point average of all successful students on a 10-point scale) or (Mean of the percentage of marks of all successful students in 2rd year/10)	7.71	7.56	7.54
Y=Total no. of successful students	46.00	45.00	43.00
Z=Total no. of students appeared in the examination	49.00	45.00	47.00
API [X * (Y/Z)]	7.24	7.56	6.90

Average API [(AP1 + AP2 + AP3)/3] : 7.23

B8. Academic Performance of the Third Year Students of the Program

Table No.B8.1: Academic Performance of the Third Year Students of the Program

Academic Performance	CAYm1 (2024-25)	CAYm2 (2023-24)	CAYm3 (2022-23)
X=(Mean of 3rd year grade point average of all successful students on a 10-point scale) or (Mean of the percentage of marks of all successful students in 3rd year/10)	7.81	7.85	7.91
Y=Total no. of successful students	45.00	43.00	49.00
Z=Total no. of students appeared in the examination	45.00	43.00	49.00
API [X*(Y/Z)]:	7.81	7.85	7.91

Average API [(AP1 + AP2 + AP3)/3] : 7.86

B9. Placement, Higher Studies, and Entrepreneurship

Table No.B9.1: Placement, higher studies, and entrepreneurship details.

Item	LYG (2021-22)	LYGm1(2020-21)	LYGm2(2019-20)
FS*=Total no. of final year students	61.00	60.00	60.00
X=No. of students placed	19.00	23.00	24.00
Y=No. of students admitted to higher studies	6.00	18.00	19.00
Z= No. of students taking up entrepreneurship	0.00	0.00	0.00
Placement Index(P) = $((X + Y + Z)/FS) * 100$:	40.98	68.33	71.67

Average Placement Index = (P_1 + P_2 + P_3)/3: 60.33 Placement Index Points:

PART C: Faculty Details in Department and Allied Departments**(Data to be filled in for the Department and Allied Departments)****C1. Faculty details of Department and Allied Departments**

Table No.C1: Faculty details in the Department for the past 3 years including CAY

Sr.No	Name of the Faculty	PAN No.	Highest degree	University	Area of Specialization	Date of Joining in this Institution	Experience in years in current institute	Designation at Time Joining in this Institution	Present Designation	The date on which Designated as Professor/ Associate Professor if any	Nature of Association (Regular/ Contract/ Ad hoc)	Currently Associated (Y/N)	In case of NO, Date of Leaving	IS HOD?
1	Dr. B. S. Hungund	XXXXXXX62A	Ph.D	Dr. B. A. Marathwada University, Aurangabad	Microbiology	25/08/2003	22.8	Lecturer	Professor	01/11/2011	Regular	Yes		Yes
2	Dr. Uday M Muddapur	XXXXXXX43M	Ph.D	Karnataka university Dharwad	Biochemistry	21/07/2016	9.9	Professor	Professor	21/07/2016	Regular	Yes		No
3	Dr. Zabin K Bagewadi	XXXXXXX12F	Ph.D	Karnataka University, Dharwad	Biochemistry	07/03/2006	20.2	Lecturer	Professor	01/08/2025	Regular	Yes		No
4	Dr. L. R. Patil	XXXXXXX25F	Ph.D	Visvesvaraya Technological University, Belagavi	Biochemical Engineering	25/08/2003	22.8	Lecturer	Associate Professor	01/03/2013	Regular	Yes		No
5	Dr.V. S. Hombalimath	XXXXXXX20E	Ph.D	Visvesvaraya Technological University, Belagavi	Biochemical Engineering	01/08/2005	20.9	Lecturer	Associate Professor	01/03/2013	Regular	Yes		No
6	Dr. S. V. Desai	XXXXXXX03P	Ph.D	University of Mysore	Food Microbiology	02/08/2013	12.9	Assistant Professor	Associate Professor	01/07/2017	Regular	Yes		No
7	Dr. Anil R. Shet	XXXXXXX47Q	Ph.D	KLE Technological University, Hubballi	Biochemical Engineering	15/03/2004	22.2	Lecturer	Assistant Professor		Regular	Yes		No
8	Dr. Gururaj B. Tennalli	XXXXXXX56L	Ph.D	KLE Technological University, Hubballi	Biotechnology	11/08/2008	17.9	Lecturer	Assistant Professor		Regular	Yes		No

9	Dr. Deepak A. Yaraguppi	XXXXXXXX94P	Ph.D	KLE Technological University, Hubballi	Biotechnology	10/07/2010	15.10	Lecturer	Assistant Professor		Regular	Yes		No
10	Dr. Sharanappa A	XXXXXXXX92L	Ph.D	KLE Technological University, Hubballi	Computational Biology	23/08/2013	12.8	Assistant Professor	Assistant Professor		Regular	Yes		No
11	Dr. Mohanakrishna Gunda	XXXXXXXX79E	Ph.D	JNTU, Hyderabad.	Environmental. Biotechnology	17/01/2022	4.3	Associate Professor	Associate Professor	17/01/2022	Regular	Yes		No

Table No.C2: Faculty details of Allied Departments for the past 3 years including CAY.

C2. Student-Faculty Ratio (SFR)

No. of UG(Engineering) programs in Department including allied departments/ clusters (UGn):

UG1=1st UG program

UGn=nth UG program

B= No. of Students in UG 2nd year (ST)

C= No. of Students in UG 3rd year (ST)

D= No. of Students in UG 4th year (ST)

No. of PG (Engineering) programs in Department including allied departments/ clusters (PGm):

PG1=1st PG program.

PGm=mth PG program

A= No. of Students in PG 1st year

B= No. of Students in PG 2nd year

Student Faculty Ratio (**SFR**) = S/F

S= No. of students of all programs in the Department including all students of allied departments/clusters.

No. of students (ST)=Sanctioned Intake (SA)+ Actual admitted students via lateral entry including leftover seats (L) if any (limited to 10 % of SA)

Students who admitted under supernumerary quotas (SNQ, EWS, etc) will not be considered in calculating SFR value. Those students are exempted.

F=Total no. of regular or contractual faculty members (Full Time) in the Department, including allied departments/clusters (excluding first year faculty (The faculty members who have a 100% teaching load in the first-year courses)).

No. of UG Programs in the Department1 No. of PG Programs in the Department0

Table No.C2.1: Student-faculty ratio.

Description	CAY(2025-26)	CAYm1 (2024-25)	CAYm2 (2023-24)
UG1.B	61	62	62
UG1.C	62	61	61
UG1.D	60	60	60
UG1: Biotechnology	183	183	183
DS=Total no. of students in all UG and PG programs in the Department	183	183	183
AS=Total no. of students of all UG and PG programs in allied departments	0	0	0
S=Total no. of students in the Department (DS) and allied departments (AS)	S1= 183	S2= 183	S3= 183
DF=Total no. of faculty members in the Department	11	11	11
AF= Total no. of faculty members in the allied Departments	0	0	0
F=Total no. of faculty members in the Department (DF) and allied Departments (AF)	F1= 11	F2= 11	F3= 11
FF=The faculty members in F who have a 100% teaching load in the first-year courses	0	0	0
Student Faculty Ratio (SFR)=S/(F-FF)	SFR1= 16.64	SFR2= 16.64	SFR3= 16.64
Average SFR for 3 years	SFR= 16.64		

C3. Faculty Qualification

- Faculty qualification index (FQI) = 2.5 * [(10X +4Y)/RF] where
- X=No. of faculty members with Ph.D. degree or equivalent as per AICTE/UGC norms.
- Y=No. of faculty members with M. Tech. or ME degree or equivalent as per AICTE/ UGC norms.
- RF=No. of required faculty in the Department including allied Departments to adhere to the 20:1 Student-Faculty ratio, with calculations based on both student numbers and faculty requirements as per section C2 of this documents: (RF=S/20).

Table No.C3.1: Faculty qualification.

Year	X	Y	RF	FQ = 2.5 x [(10X + 4Y) / RF]
2025-26(CAY)	11	0	9.00	30.56
2024-25(CAYm1)	8	3	9.00	25.56
2023-24(CAYm2)	8	3	9.00	25.56

C4. Faculty Cadre Proportion

- Faculty Cadre Proportion is 1(RF1): 2(RF2): 6(RF3)
- RF1= No. of Professors required = 1/9 * No. of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (S) as per C2 of this documents:.
- RF2= No. of Associate Professors required = 2/9 * No. of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (S) as per section C2 of this documents:.
- RF3= No. of Assistant Professors required = 6/9 * No. of Faculty required to comply with 20:1 Student-Faculty ratio based on no. of students (S) as per section C2 of this documents:.
- Faculty cadre and qualification and experience should be as per AICTE/UGC norms.

Table No.C4.1: Faculty cadre proportion details.

Year	Professors		Associate Professors		Assistant Professors	
	Required RF1	Available AF1	Required RF2	Available AF1	Required RF3	Available AF3
2025-26	1.00	3.00	2.00	4.00	6.00	4.00
2024-25	1.00	2.00	2.00	5.00	6.00	4.00
2023-24	1.00	2.00	2.00	5.00	6.00	4.00
Average	RF1=1.00	AF1=2.33	RF2=2.00	AF2=4.67	RF2=6.00	AF2=4.00

C5. Visiting/Adjunct Faculty/Professor of Practice

Table No. C5.1: List of visiting/adjunct faculty/professor of practice and their teaching and practical loads.

(CAYm1)

(CAYm2)

(CAYm3)

C6. Academic Research

Table No. C6.1: Faculty publication details.

S.No.	Item	2024-25 (CAYm1)	2023-24 (CAYm2)	2022-23 (CAYm3)
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1	No. of peer reviewed journal papers published	32	28	21
2	No. of peer reviewed conference papers published	0	1	2
3	No. of books/book chapters published	2	2	0

C7. Sponsored Research Project

Table No. C7.1: List of sponsored research projects received from external agencies.

(CAYm1)

(CAYm2)

(CAYm3)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
Dr. Gunda Mohanakrisna	-	Biotechnology (School of advanced Sciences)	Microbial electrosynthesis based production of ethylene from the valorization of food and food processing waste	SERB	3 Years	46.01
Dr. S.V. Desai	Dr. V. S. Hombalimath	Biotechnology	Nutritional Profile and Metagenomics Insights of Microbial Diversity of Nuchhu Ambli- a millet-based traditional Fermented Food of North Karnataka.	VGST	1	7.50
						Amount received (Rs.):53.51

Total Amount (Lacs) Received for the Past 3 Years: 53.51**Note*:**

- Only sponsored research projects will be considered. Infrastructure-based projects will not be considered here.

C8. Consultancy Work

Table No. C8.1: List of consultancy projects received from external agencies.

(CAYm1)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
						Amount received (Rs.):0

(CAYm2)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
						Amount received (Rs.):0

(CAYm3)

PI Name	Co-PI names if any	Name of the Dept., where project is sanctioned	Project Title*	Name of the Funding agency	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25
						Amount received (Rs.):0

Total amount (Lacs) received for the past 3 years: 0

Note*:

- Only consultancy projects will be considered. Infrastructure-based projects will not be considered here.

C9. Institution Seed Money or Internal Research Grant to its Faculty for Research Work

Table No. C9.1: List of faculty members received seed money or internal research grant from the Institution.

(CAYm1)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Dr. G. B Tennalli	Capacity Building	1 YEAR	2.00	1.50	PhD
DR. B. S Hungund	RG Group Grant	1 Year	5.00	4.81	PhD and Publication
			Amount received (Rs.): 7.00		

(CAYm2)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Dr. Uday M Muddapur	Capacity Building	1 Year	0.80	0.80	1 PhD and 1 paper
Dr. Zabin K Bagewadi	Capacity Building	1 Year	1.30	1.30	1 PhD and 2 paper
Dr. L R Patil	Capacity Building	1 Year	0.70	0.70	1 Paper
			Amount received (Rs.): 2.80		

(CAYm3)

Faculty name	Project title/ Support for Activity	Duration of the project	Amount(Lacs) i.e. 15,25,000=15.25	Amount Utilized(Lacs) i.e. 15,25,000=15.25	Outcomes of the project
Dr. Shivaprakash Hiremath	Molecular expression	1 year	4.40	4.35	Publications
			Amount received (Rs.): 4.40		

Total amount (Lacs) received for the past 3 years : 14.20

PART D: Laboratory Infrastructure in the Department
(Data to be filled in for the Department)

D1. Adequate and Well-Equipped Laboratories, and Technical Manpower

Table No.D1.1: List of laboratories and technical manpower.

Sr. No	Name of the Laboratory	Number of students per set up(Batch Size)	Name of the Important Equipment	Weekly utilization status(all the courses for which the lab is utilized)	Technical Manpower Support		
					Name of the Technical staff	Designation	Qualification
1	Biochemistry Lab (BTL-1)	15	1.HPLC 2.Thermostatic water bath 3.Digital Photo Colorimeter, 4.Orbital shaker, 5.Laminar Air Flow	25	Basavaraj Aralikatti	Assistant Instructor	B.Sc.
2	Bioprocess Engineering Lab (BTL-2)	15	1.Fermenter, 2. Cooling centrifuge, 3. Ultra-Sonicator Homogenizer, 4. Electronic orbital shaker	30	Basavaraj Aralikatti	Assistant Instructor	B.Sc.
3	Cell & Molecular Lab(BTL-3)	15	1.Thermo cycler, 2. Gel Documentation system 3. Mini-Centrifuge, , 4. UV-Visible spectrophotometer	20	S.N. Joshi	Instructor	M.Sc.
4	Bioinformatics (BTL-4)	15	Desktop Computer systems (17 No's)	12	Mahantesh Benakanalma	Instructor	Diploma (CS)
5	Microbiology (BTL-5)	15	1.Freeze dryer 2. Phase Contrast Microscope 3. Laminar air flow 4. Orbital Shaker 5. Cold centrifuge	12	S.N. Joshi	Instructor	M.Sc.
6	Unit Operation (BTL-6)	15	1.Mixed flow reactor, 2. Plug flow reactor 3. Vertical condenser, 4. Heat transfer in packed bed, Wetted	25	Jayadevi Ammanagi	Assistant Instructor	BCA PGDCA
7	Process Control Lab (BTL-7)	15	1.Temperature Control Trainer 2. Interacting & Non-Interacting Systems 3. Pressure Control Trainer, 4.	20	Jayadevi Ammanagi	Assistant Instructor	BCA PGDCA

D2. Safety Measures in Laboratories

Table No. D2.1: List of various safety measures in laboratories.

Sr. No	Laboratory Name	Safety Measures
1	Biochemistry Lab (BTL-1)	1. Fire extinguisher 2. First Aid Box. 3. Display of emergency contact numbers. 4. Display of general safety guidelines and management of hazardous chemicals. 5. Material Safety Data Sheet (MSDS) maintained. 6. Use of personal protective equipments (PPE) like goggles, gloves etc.
2	Bioprocess Lab (BTL-2)	1. Fire extinguisher 2. First Aid Box 3. Display of emergency contact numbers. 4. Procedure for disposal of waste is defined, displayed and followed. 5. Display of general safety guideline
3	Cell & molecular Lab (BTL-3)	1. Fire extinguisher. 2. Display of emergency contact numbers. 3. Material Safety Data Sheet (MSDS) maintained 4. Procedure for disposal of waste is defined, displayed and followed. 5. Use of personal protective equipments (PPE) like goggles, hand-gloves, head caps 6. Procedure for display of Ethidium bromide is displayed and followed.
4	Bioinformatics (BTL-4)	1. Anti-virus for all the computer systems is installed. 2. Display of emergency contact numbers. 3. General guidelines to be followed are displayed.

5	Microbiology (BTL-5)	1. Fire extinguisher 2. First Aid Box 3. Display of emergency contact numbers. 4. Display of general safety guidelines. 5. Use of personal protective equipment's (PPE) like goggles, hand-gloves. 6. Procedure for disposal of waste is defined, displayed and followed.
6	Unit Operation (BTL-6)	1. Fire extinguisher 2. Display of emergency contact numbers.
7	Process Control Lab (BTL-7)	1. Display of general safety guidelines 2. Display of emergency contact numbers.
8	Faculty Research Lab- I	1. Display of general safety guidelines 2. Display of emergency contact numbers
9	Faculty Research Lab- II	1. Display of general safety guidelines 2. Display of emergency contact numbers

D3. Project Laboratory/Research Laboratory

Research Laboratory – I (Faculty and Ph.D. Scholars Research Laboratory)

Research Laboratory–I is dedicated to supporting the research activities of faculty members and doctoral scholars. The laboratory provides essential facilities for conducting advanced research in biotechnology, microbiology, bioprocessing, and related interdisciplinary areas. The laboratory is equipped with instruments such as a UV–Visible Spectrophotometer for analytical studies, an autoclave for sterilization, an incubator for microbial cell culture growth, a moisture analyzer for material characterization, a centrifuge for sample separation, and a Phase Contrast Microscope for detailed observation of biological specimens. These facilities enable researchers to undertake fundamental and applied research, generate quality publications, and contribute to funded research projects.

**Major Equipments:**

- UV–Visible Spectrophotometer
- Autoclave
- Incubator
- Moisture Analyzer
- Centrifuge
- Phase Contrast Microscope

Spectrophotometer

Centrifuge**Research Laboratory – II (Process and Product Development Laboratory)**

Research Laboratory–II is established to facilitate process optimization, scale-up studies, and product development activities. The laboratory provides advanced instrumentation for bioprocess development, downstream processing, analytical characterization, and quality assessment. It is equipped with a 50 L Pilot-Scale Fermentor for scale-up studies, HPLC for analytical and quality analysis, AKTA Protein Purification System for biomolecule purification, Microplate Reader for high-throughput analysis, Homogenizer for sample preparation, and supporting equipment such as UV–Visible Spectrophotometer, Centrifuge, pH Meter, Electronic Weighing Balance, Ultra-Pure Water System, and Water Purification System. The laboratory enables faculty, research scholars, and students to undertake translational research, process development, and industry-oriented product innovation.

**Major Equipments:**

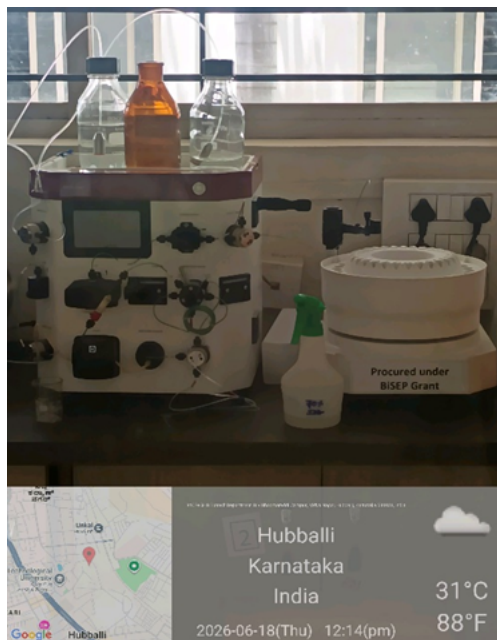
- Pilot-Scale Fermentor (50 L)
- HPLC

- Centrifuge
- AKTA Protein Purification System
- UV-Visible Spectrophotometer
- pH Meter
- Electronic Weighing Balance
- Ultra-Pure Water System
- Water Purification System
- Microplate Reader
- Homogenizer

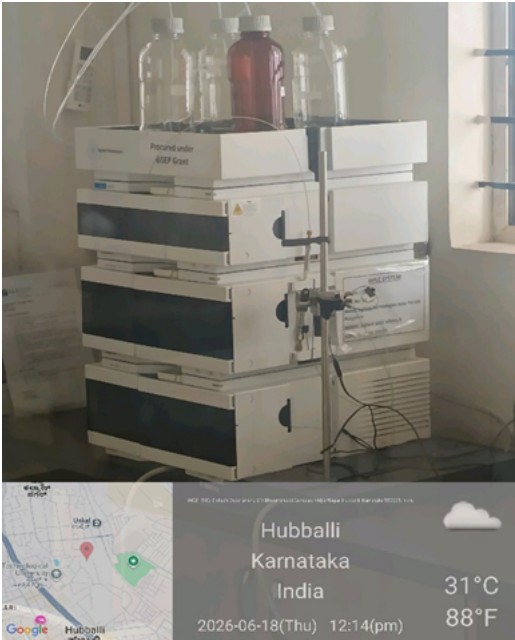
Pilot Scale Fermenter



AKTA Protein Purification System



HPLC



Spectrophotometer



Microplate Reader



External Research Funded Project

An external research project (VGST-K-FIST-L1-GRD-886) titled "Nutritional Profile and Metagenomics Insights of Microbial Diversity of *Nucchu Ambli*- a millet-based traditional Fermented Food of North Karnataka " was undertaken and completed. It was funded by Vision Group of Science and Technology, Government of Karnataka.

Memorandum of Understandings (MoU)

The department has MoUs with several biotechnology-led start-ups and regional companies with an objective to foster academic research, provide internship opportunities to students, keep abreast of the developments in the industries and also support the entrepreneurs with infrastructure and ecosystem.

The department has signed MoUs with the following companies:

1. Prajna Biosciences
2. Biosyl Technologies
3. Chemogenesis
4. Chemioplastics
5. Hosamani Nutraceuticals
6. Samarth Biotech.

Start-up incubator support:

The laboratory accommodates and supports start-up incubator companies to foster entrepreneurship through innovation and product & process development in the domain of Biotechnology. Some of the startup companies housed in the lab include WeGha Herbals, Vayodaanta Biotech Pvt. Ltd., Sanbeer Pvt. Ltd, Navya Therapeutics and others.

PART E: First Year faculty and financial Resources
(Data to be filled in for the first year course faculty and budget allocation and utilization)

E1. First Year Student-Faculty Ratio (FYSFR)

Table No. E1.1: FYSFR details.

Year	Sanctioned intake of all UG programs (S4)	No. of required faculty (RF4= S4/20)	No. of faculty members in Basic Science Courses & Humanities and Social Sciences including Management courses (NS1)	No. of faculty members in Engineering Science Courses (NS2)	Percentage= No. of faculty members ((NS1*0.8) + (NS2*0.2))/(No. of required faculty (RF4)); Percentage=((NS1*0.8) +(NS2*0.2))/RF
2023-24(CAYm2)	1380	69	45	22	60
2024-25(CAYm1)	1380	69	49	25	64
2025-26(CAY)	1800	90	56	28	56

E2. Budget Allocation, Utilization, and Public Accounting at Institute Level

Table No. E2.1: Budget and actual expenditure incurred at Institute level.

Items	Budgeted in 2025-26	Actual Expenses in 2025-26 till	Budgeted in 2024-25	Actual Expenses in 2024-25 till	Budgeted in 2023-24	Actual Expenses in 2023-24 till	Budgeted in 2022-23	Actual Expenses in 2022-23 till
Infrastructure Built-Up	275000000.00	266400332.26	292200000.00	337836058.50	320300000.00	275490296.08	226000000.00	272900679.00
Library	10800000.00	11640291.79	11600000.00	9771428.06	8100000.00	6934858.90	6550000.00	6684290.00
Laboratory equipment	77000000.00	70298745.00	40000000.00	42923569.00	30000000.00	42637137.00	60000000.00	45115063.00
Teaching and non-teaching staff salary	583700000.00	573446030.00	561500000.00	551937567.00	512840000.00	498123229.00	508167243.00	486223533.00
Outreach Programs	2000000.00	1821446.00	1700000.00	1059445.00	1200000.00	1185057.00	1800000.00	1759579.55
R&D	95000000.00	89863171.00	120000000.00	105715282.20	91000000.00	67685714.98	60500000.00	85916358.00
Training, Placement and Industry linkage	40000000.00	32995616.20	28000000.00	28096690.28	39400000.00	27868861.85	16700000.00	22804772.45
SDGs	500000.00	0	500000.00	0	0	0	0	0
Entrepreneurship	6000000.00	5500402.00	6500000.00	5765247.00	8500000.00	8288555.00	6000000.00	5721313.00
Others, specify	318247000.00	300391085.97	240450000.00	236975209.38	221575000.00	222576706.76	200300000.00	212374205.00
Total	1408247000.00	1352357120.22	1302450000.00	1320080496.42	1232915000.00	1150790416.57	1086017243.00	1139499793.00

E3. Budget Allocation, Utilization, and Public Accounting at Program Specific Level

Table No. E3.1: Budget and actual expenditure incurred at program level.

Items	Budgeted in 2025-26	Actual Expenses in 2025-26 till	Budgeted in 2024-25	Actual Expenses in 2024-25 till	Budgeted in 2023-24	Actual Expenses in 2023-24 till	Budgeted in 2022-23	Actual Expenses in 2022-23 till
Laboratory equipment	230000.00	389119.00	200000.00	169099.00	25000.00	20768.00	400000.00	325632.00
Software	140000.00	0	0	0	0	0	0	0
SDGs	50000.00	0	50000.00	0	0	0	0	0
Support for faculty development	200000.00	176399.00	100000.00	72999.00	450000.00	351688.00	110000.00	108637.00
R & D	4050000.00	2827447.00	1500000.00	1502625.00	1000000.00	1079159.00	450000.00	373263.00
Industrial Training, Industry expert, Internship	500000.00	375140.00	600000.00	575725.00	550000.00	388186.00	690000.00	654500.00
Miscellaneous Expenses*	40395000.00	38540945.20	33400000.00	33184165.38	54180000.00	52204453.23	44597000.00	45283435.00
Total	45565000.00	42309050.20	35850000.00	35504613.38	56205000.00	54044254.23	46247000.00	46745467.00